

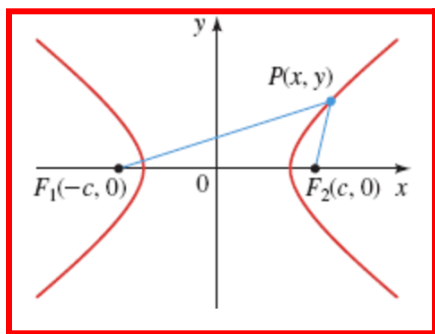
Chp 11.3: Hyperbolas

Although ellipses and hyperbolas have completely different shapes, their definitions and equations are similar. Instead of using the **sum** of distances from two fixed foci, as in the case of an ellipse, we use the **difference** to define a **hyperbola**.

Geometric Definition of a Hyperbola

A **hyperbola** is the set of all points in the plane, the **difference** of whose **distances** from **two** fixed points F_1 and F_2 is a **constant**. These two fixed points are the **foci** of the hyperbola.

P is on the hyperbola if $|d(P, F_1) - d(P, F_2)| = 2a$.



The **standard equation** of a hyperbola with foci on the x -axis and center at the origin is

$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. If we replace x by $-x$ or y by $-y$ in this equation, it remains unchanged, so the

hyperbola is symmetric about **both the x - and y -axes and about the origin**. The **x -intercepts** are **$\pm a$** , and the points **$(a, 0)$ and $(-a, 0)$** are the **vertices** of the hyperbola. There is no

y -intercept, because setting $x = 0$ in the equation of the hyperbola leads to $-y^2 = b^2$ which has no real solution. The hyperbola consists of **two parts**, called its **branches**. The segment joining the two vertices on the separate branches is the **transverse axis** of the hyperbola, and the origin is called its **center**.

If we place the foci of the hyperbola on the y -axis rather than on the x -axis, this has the effect of **reversing the roles of x and y** in the derivation of the equation of the hyperbola. This leads to a hyperbola with a vertical transverse axis.

Hyperbola with Center at the Origin

The graph of each equation is a hyperbola with center at the origin and having the given properties.

EQUATION

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \quad (a > 0, b > 0) \quad \frac{y^2}{a^2} - \frac{x^2}{b^2} = 1 \quad (a > 0, b > 0)$$

VERTICES

$$(\pm a, 0)$$

$$(0, \pm a)$$

TRANSVERSE AXIS

Horizontal, length $2a$

Vertical, length $2a$

ASYMPTOTES

$$y = \pm \frac{b}{a}x$$

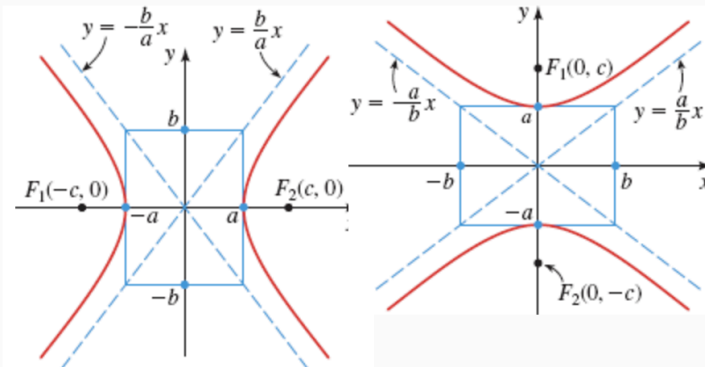
$$y = \pm \frac{a}{b}x$$

FOCI

$$(\pm c, 0), c^2 = a^2 + b^2$$

$$(0, \pm c), c^2 = a^2 + b^2$$

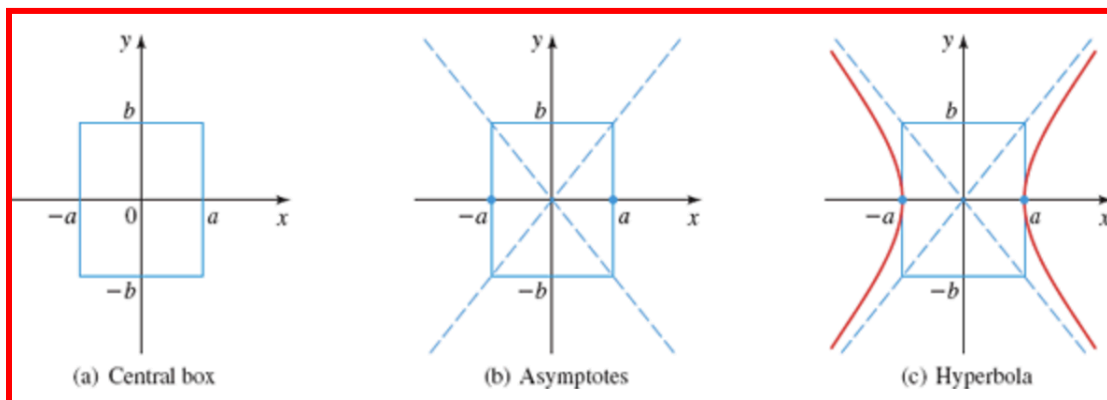
GRAPH



How to Sketch a Hyperbola

1. Sketch the **Central Box**. This is the rectangle centered at the **origin**, with sides parallel to the axes, that crosses one axis at $\pm a$ and the other at $\pm b$.
2. Sketch the **Asymptotes**. These are the lines obtained by **extending the diagonals** of the central box.
3. Plot the **Vertices**. These are the **two x-intercepts** or the **two y-intercepts**.
4. Sketch the **Hyperbola**. Start at a **vertex**, and sketch a branch of the **hyperbola**, approaching the **asymptotes**. Sketch the other branch in the same way.

Steps in graphing the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.



Example 1: An equation of a hyperbola is given.

(a) Find the vertices, foci, and asymptotes of the hyperbola.

(b) Determine the length of the transverse axis.

(c) Sketch a graph of the hyperbola.

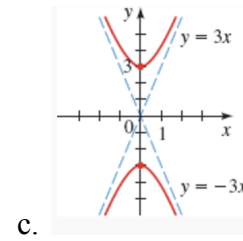
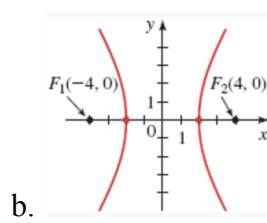
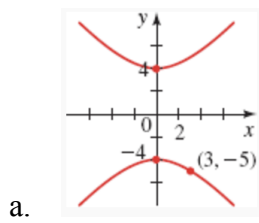
i. $\frac{x^2}{9} - \frac{y^2}{64} = 1$

ii. $\frac{y^2}{36} - \frac{x^2}{4} = 1$

iii. $x^2 - 3y^2 + 12 = 0$

iv. $9x^2 - 16y^2 = 1$

Example 2: Find the standard equation for the hyperbola whose graph is shown.



Example 3: Find the standard equation for the hyperbola that satisfies the given conditions.

a. Foci: $(\pm 6, 0)$ vertices: $(\pm 2, 0)$

b. Vertices: $(\pm 1, 0)$, asymptotes: $y = \pm 5x$

c. Asymptotes: $y = \pm x$, hyperbola passes through $(5, 3)$

d. Foci: $(0, \pm 1)$, length of transverse axis: 1